

MMR-OS

Operating System Concepts (10th Edition) (Dino on cover)

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OS → Operating System
↳ Firmware

Practical → Linux Installation

↳ CLI-based

↳ basic command, operations

↳ Multi-threading

↳ (concurrency)

↳ Thread-Library

Resource:

Partha Bhakta

↳ Project (Linux and Windows

[Robin: Payment Gateway
for CSE/CSIT]

Server Installation and
Configuration)

- Code

- Algorithm

- Automation Knowledge

- Hosting, Reporting

- Database

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Introduction - Slide [611 - 27]

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Slides: - 28, 29, 30, 31, 32, 33

Transient Space → Operating System

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Slides: 34 - 51

Dual-Mode Operation

Transition from User to kernel mode

Timer

→ (!) Chapter-6. Deadlock

Process Management

↳ Program Counter

Process Management Activities

Memory Management

File-system Management

Mass-storage Management

Caching

Characteristics of various types of storage

Migration of Data from A to

I/O Subsystem (terms & & &)

Protection and Security

Virtualization

(!) Java and Python
Concurrency
and Multithreading

Computing Environments - Virtualization

Distributed Systems (1) Windows Server

Setup

Grouping on a distributed
network

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1. Lab work - 1 : Basic Linux and Windows
Commands

2. Lab task - 2 : Virtualization

3. Lab task - 3 : Creating network ~~environment~~

Windows Server Setup

↳ build network
and exchanged data by

CAN .

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Slides: 52 - last

Computer-System Architecture

Symmetric Multiprocessing

Dual-Core Design

Non-uniform Memory Access System

Clustered Systems (1) SAN - Storage

Area Network

Client Server

(!) High Performance

Cloud Computing

Computing - HPC

Real-time Embedded Systems

Free and Open-Source Operating Systems

Kernel Data Structures

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Chapter 2: Operating-System Services

Sides: 1 - 32

ISDN

~~Error~~ Parity, CRC,

Registry Editor

{ Lab work:
Use Registry Editor for Digital Forensics

① Side Channel Attack

A - Authentication

A - Authorization

A - Accountability

Confidentiality, Integrity, Availability

System Calls , Types of System Calls

slide ~~2.15~~ 2.15

Lab work 2.24 (slide)

→ System calls implementations

- Python or Java (if possible)

- otherwise C/C++.

2.26, 2.27,

HW: Linkers and loaders

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slides: 32 -

Linkers and Loaders ① DDL - Dynamically Linked Libraries

ABI → Application Binary Interface

user goals and system goals ~~Linux~~ [Linux]

Slide 38 details

Darwin kernel - study

① Aqua ② OS, Cocoa - programming environment

Slide - 53

Lab task: build a Linux kernel.

System Boot

UEFI → Unified Extensible Firmware Interface

* Kernighan's Law:

Tracing - Slide: 57

Lab task: Slide: 59

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OS → Book

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Chapter 03 - 0

Slides: 1 - 12

process concept

Memory layout of a C program

process state (5 states)

Process control block (PCB)

↳ ~~Resource~~ RSVP (RFC)

Chapter 5 → Round Robin and other algorithms

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Chapter. 03 -

Slides: 3.11 - 3.27

Threads

Process Scheduling

↳ dispatcher (program)

↳ Goal - Maximum CPU utilization

- ↳ Scheduling queues
 - ↳ ready queue (in memory)
 - ↳ waiting queue (event eg. I/O)

Ready and waiting queue

Representation of process scheduling

CPU switch from Process to Process

- ↳ PCB - Process Control Block

Context switch

- ↳ overhead (time between switching)
time waste

Multitasking in Mobile Systems

Operations on processes

Process Creation

- ↳ Unix Example

`fork()`, `exec()`, `wait()`

① A tree of processes in Linux

~ Unix
~ windows API

Process Termination

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Chapter 3

↳ Slides: 3.28 -

Multiprocess Architecture - Chrome Browser

Interprocess Communication

↳ IPC → Interprocess Communication

Communications Models

↳ a) Shared memory

↳ b) Message passing

Producer-Consumer Problem

↳ Paradigm → FIFO

IPC - Shared Memory

Bounded-Buffer - Shared Memory

Producer Process - Shared Memory

Consumer ~ ~ ~ (Selfs)

What are filling all the buffer

Race Condition

IPC - Message Passing

Message Passing (cont.)

Implementation of Communication Link

Direct Communication

Indirect Communication (cont.)

Buffering

IPS - POSIX (Self Study)

Mail message (Linux)

reciprocal

Local Procedure call (windows)

Advanced Local Procedure call (LPC)

Pipes

↳ one like inheritance other is line interfacing

Ordinary pipes

↳ Anon.

Named pipes

Socket

↳ Lab Work: Java Socket Programming
Chatting (App). between two programs.

Remote Procedure calls

↳ MIDL - Microsoft Interface Definition Language

Execution of RPC

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Chapter 4: Threads and Concurrency

↳ Slides: 1-4, 4-51

(!) Process creation is heavy-weight ~~and~~ while
threads creation is light-weight.

Multithreaded Server Architecture

Benefits

↳ ~~Responsive~~ Responsiveness

↳ Resource sharing

↳ Economy

↳ Scalability

Multicore Programming

↳ Multicore or multiprocessor

Challenges

↳ Parallelism

↳

Concurrency vs Multicore

Multicore Programming

↳ Data Parallelism

↳ Task Parallelism

Data and Task Parallelism

Amdahl's Law

↳ S is serial portion

↳ N processing cores

$$\text{speedup} \leq \frac{1}{S + \frac{(1-S)}{N}}$$

User Threads and Kernel Threads

Multithreading Models.

Many-to-one

One-to-one

Many-to-Many

Two-level Model

Thread Libraries

Pthreads

Java Threads

Implicit Threading }

Thread pools

Fork-Join Parallelism ~~+~~ in Java

Grand Central Dispatch

Intel Threading Building Blocks (TBB)

Threading Issues

Semantics of `fork()` and `exec()`

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Slide: 51 - End

Signal Handling

2 signal handlers

↳ default

↳ user-defined

Thread-Local Storage (TLS)

6 Lightweight Processes

Windows Threads

Linux Threads

Chapter 5 : CPU Scheduling

Slide: 1 - 67

Dispatcher / Scheduler

Preemptive and Nonpreemptive Scheduling

Dispatcher.

↳ dispatcher latency.

Scheduling criteria

* ↳ definitions

Scheduling Algorithm - Optimization criteria

FCFS → scheduling

↳ convoy effect

Shortest-Job-First ^(SJF) Scheduling

Determining length of next CPU Burst

Shortest-Remaining-time-first

Round-Robin (RR)

↳ Arrival Time

* Priority Scheduling

↳ Arrival and waiting time

* ~~Priority~~ ~~Scheduling~~ w/ Round Robin

Multi-level Queue

Thread Scheduling

Multiple-core $\rightarrow$ Multiple Process scheduling.

Multitimed and multicore

to Rate ~~vs~~